

Active Inference as a Process Theory of Case

Static Categories Are Not Enough

The preceding sections constructed a mathematical infrastructure for analyzing case systems—categorical, type-logical, distributional, enriched, and topos-theoretic. Yet these frameworks remain *static*: they describe the structure of case grammar without explaining how a cognitive agent *deploys* that structure during real-time comprehension and production. Bridging this gap requires a dynamic *process theory* of case-marked relational reasoning.

Active inference [friston2017active] provides exactly this missing dynamic computational layer.

Surprise Minimization as Case Parsing

Free Energy Bounds Surprisal

Active inference is the primary process theory derived from the free energy principle (FEP): every self-organizing system maintains its structural integrity by minimizing the surprisal (negative log-probability) of its sensory observations under an internal *generative model* of its environment [Friston2010free]. The system executes this minimization through two complementary strategies:

1. **Perceptual inference:** Update internal beliefs to better predict current observations (reduce prediction error)
2. **Active inference:** Act on the environment to bring observations in line with predictions (reduce expected prediction error)

Recent extensions of active inference to linguistics and cognitive science have modeled language comprehension and production as forms of sequential Bayesian inference. As Donnarumma, Frosolone, and Pezzulo (2023) demonstrate in their integration of large language models and active inference for modelling eye